Transparency, Accountability and Human Oversight

1. The black-box problem
Deep neural networks are often opaque: even their developers cannot fully explain individual predictions. In domains such as credit, healthcare and criminal justice, people affected by a decision deserve an explanation they can contest. Explainability techniques such as SHAP, LIME and counterfactual explanations offer partial insight but can be unfaithful to the model's real reasoning.

2. Accountability gaps
When an AI system causes harm, responsibility is diffuse across data providers, developers, deployers and users. Clear lines of accountability should be assigned before deployment. Model cards and datasheets for datasets document intended use, limitations and evaluation results, helping downstream users judge suitability.

3. Human oversight
Human-in-the-loop review is essential for high-stakes decisions, but oversight is only effective if reviewers have the time, training and authority to override the system. Automation bias, where people over-trust machine output, can make nominal oversight meaningless. Incident reporting and redress mechanisms should be available to affected individuals.